**De Montfort University**
**Beng (Hons) Mechanical Engineering – Year 3**
**Module:** Advanced Mechanical Materials
**Experiment:** Thin Film Deposition Techniques – Spin Coating and PECVD

# Lab Report: Thin Film Deposition on Silicon Wafer Using Spin Coating and PECVD

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**Abstract – This experiment investigates the effects of deposition parameters on thin films and uniformity using two techniques: coating the substrate in polyvinyl acetate (PVAc) using spin coating, and depositing diamond like carbon on the substrate using Plasma-enhanced chemical vapour deposition. Spin-Coating is a manual process where perfect replicability is not realistically possible however, variables can be modified to minimise error. PECVD in contrast, is superior in film quality, the process is highly replicable through the techniques system. To conclude, PECVD is superior in industrial environments where uniformity, accuracy and exact replication is key.**

**Terminology - Gaseous Precursor, Physical, Chemical, Static, Thermal Expansion Coefficient, sccm, mTorr**

# I. Introduction

Thin film deposition is essential in semiconductors, solar cells, optics and other applications which require nanometre precision and uniformity [1]. Two common techniques are Spin coating, a **physical** deposition method relying on centrifugal force to spread a liquid film, and PECVD, a **chemical** technique that relies on plasma-assisted reactions to form a solid layer on the substrate.

In spin coating, a liquid solution is administered on a rotating substrate at high angular velocity, *1000 rpm – 7000 rpm,* in this experiment. The consequent film thickness is predominantly dictated by fluid viscosity, spin speed, and solvent evaporation. According to the Meyerhofer model [2]: the thickness of the film is inversely proportional to the square of spin speed (*f*), and directly proportional to both initial viscosity ($v_0$) and evaporation rate (*e*). In its simplified form, this relationship can be expressed as shown in *(1)*.

*(1). simplified Meyerhofer model for empirical understanding:* $h \propto \frac{v_0 e}{f^2}$

PECVD is the decomposition of gaseous precursors using **plasma** for deposition at lower temperatures than CVD [3].

The objectives of this study are to:

1. Investigate the effects of solution concentration and spin speed on film thickness and uniformity.
2. Compare the results to PECVD in terms of thickness control and surface uniformity.
3. Understand the working principles behind ellipsometry.

# II. Methodology

In this experiment PECVD is used to deposit a film of Diamond like Carbon onto a steel substrate. Ellipsometry cannot be used to determine the film thickness on the steel substrate. This is because this materials surface roughness is too high for the ellipsometer since the laser requires an optically flat surface to reflect on. Due to this, a silicon substrate (prepared using the same method as spin-coating), is used collaboratively since the surface properties of silicon wafers are adequate for reliable data analysis using ellipsometry.

*Equipment:* Ultrasonic bath, Diamond Scribe, Nitrogen gun, Beaker and a pair of Tweezers.

*Consumables:* Silicon Wafers, Steel Substrate, Sample box, Glass, Acetone, Methanol, IPA and DI water.

*Personal Protective Equipment (PPE):* Lab coat, Gloves, Safety glasses.

*Cleaving the substrate* - the substrate is a silicon wafer of **p (100)** crystal orientation. The wafer was sectioned into pieces adequate in size using a diamond scribe, to fit correctly in the spin-coater.

*Cleansing of the substrate* – to ensure the best uniformity and accuracy of the resulting thin film, impurities should be removed. the substrates were placed in solvents; Acetone, Methanol and Isopropyl Alcohol (IPA) to remove organic impurities followed by De-Ionised water to remove solvent residue. The substrates were cleaned in these solutions in conjunction with the ultrasonic bath for *5 min* to enhance the removal of particulate contaminants. The nitrogen gun was then used until dry to prevent recontamination and water spots. Finally, the substrates were placed in a clean box after cleaning to prevent recontamination.

*Note: for PECVD the steel substrate was cleaned similarly as the silicon wafers; DI water was not used as it facilitates rust which can also dissolve, leading to defects and potentially contaminating the PECVD chamber.*

## A. Spin-Coating Process

*Equipment:* Spin Coater, Ultrasonic bath, Vials, Ellipsometer, Analytical Balance, Spatula and a pair of Tweezers.

*Consumables:* Silicon Wafers, Sample box, Solution Vials, PVAc and Methanol.

*Personal Protective Equipment (PPE):* Lab coat, Gloves, Safety glasses.

*Solution Preparation* – the PVAc polymer pellets were weighed. Methanol was added to dissolve the polymer. Sonication agitates the polymer particles, accelerating the solvation process. The solutions were filtered to significantly reduce impurities and any undissolved polymer. Two solutions of *10mg/ml* and *20mg/ml* were prepared to demonstrate the effects of viscosity in spin coating.

*Spin-Coating & Analysis* – thin films were applied at *1000-7000 rpm* in *1000 rpm* increments through **static** spin coating to display the effect of spin speed to film thickness.

Ellipsometry was used to measure film thickness at the centre for all substrates, at the corner for the lowest and highest RPM and, on the sides of the *20 mg/ml* substrates. This technique is non-invasive and provides nanometre precision.

## B. PECVD

*Equipment: PECVD system*
*Consumables: Steel Substrate, Silicon Wafer, Methane, Argon*
*Personal Protective Equipment (PPE): -*

*Experimental Setup* – gaseous precursors are controlled and connected to a chamber. The chamber is connected to a vacuum pump and withholds a powered cathode and electrode of *40mm* distance from each other in combination with an applied Radio Frequency (RF) of *13.56MHz* to generate and control plasma for thin film deposition.

The use of plasma allows for PECVD to be performed at relatively low temperatures *(room temp to 400°C)*, compared to CVD *(600°C-800°C)*. lower operating temperatures reduce the energy consumption of the process and mitigate risk of

**thermal expansion coefficient** differences (between substrate and film) resulting in cracking. Electrodes being closer together is advantageous because the plasma is more uniformly dense due to a more uniformly strong electric field. When too far apart the decay of gaseous precursors loses homogeneity, leading to variations in deposition rate and film properties.

A vacuum is used to reduce potential collisions. Collisions scatter reactant particles, reduce uniformity and can deposit undesired material onto the substrate.

Coolant is used to prevent localised heating, potentially damaging sensitive components. There is no need for thermal excitation since the deposition is effective at room temperature.

*Deposition Process* – Both the steel and silicon substrate were placed on the lower electrode. The vacuum was cycled three times to ensure no residual gases remain; remaining gases can react with introduced gases, producing unwanted and potentially dangerous outcomes. Argon and Methane were introduced into the chamber at *100 sccm* and *10 sccm* respectively; at a pressure of *100 mTorr*. RF power is turned on at *100 Watts*. After *10 min* the substrate is removed.

*Analysis* – Consequently, film thickness is measured using the ellipsometer at *50* different points on the substrate; measuring the film thickness at numerous different points means derived uniformity is relatively accurate.

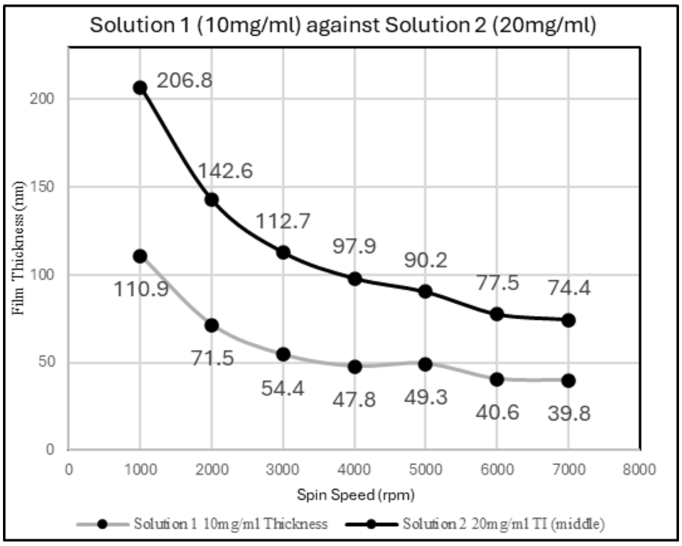
## III. Results and discussion

### A. Spin-Coating

Emslie-Bonner-Peck model [4]. where film thickness ($h$) is inversely proportional to spin speed *(ω)* square rooted, presented as (2). Applicable for non-volatile Newtonian fluids.

$$(2)\ h \propto \frac{1}{\omega^{1/2}}.$$

*Fig. 1. – Effect of solution concentration on film thickness.*




$\boldsymbol{\mu_1} = 59.2\ nm,\ \boldsymbol{\mu_2} = 114.6\ nm,\ \boldsymbol{\sigma_1} = 25.2\ nm,\ \boldsymbol{\sigma_2} = 46.9\ nm,\ \boldsymbol{RSD_1} = 42.6\%,\ \boldsymbol{RSD_2} = 40.9\%$

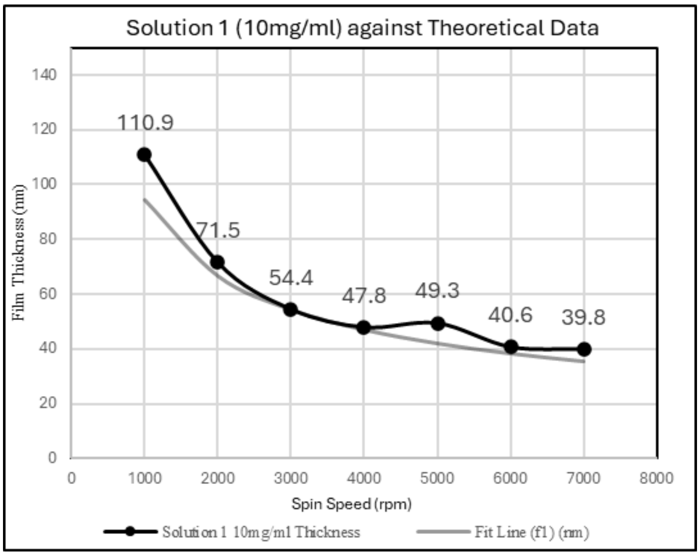
*Fig. 1* Demonstrates the significance of viscosity in determining film thickness e.g. the difference between $\mu_1$ and $\mu_2$ *(93.6%)*. This approximately follows the empirically simplified Meyerhofer's law (1) furthermore, the equation that can be derived from it as (3) $h = k\frac{v_0 e}{f^2}$, although not exactly. The reason the results are not exactly proportional, can be attributed to experimental errors. Errors such as dispensing varied amounts of solution for each substrate used, and non-homogenous substrate shapes among other reasons.

Variations such as the anomalous result at *5000 rpm* has potential causes: incomplete evaporation, edge bead effects (build-up of solution around the edge of the substrate) and improper reflection of ellipsometer laser due to lack of uniformity.

For both Fig.1 and Fig. 2 a *(k)* value for Emslie's law (2) has been calculated for both solutions based at *3000 rpm* to highlight experimental variables via a line of fit. Using equation (3)

$$(3)\ h = k\ \frac{1}{\omega^{1/2}};\ K_{\text{sol. 1.}} = 2979.6,\ K_{\text{sol. 2.}} = 6172.8$$

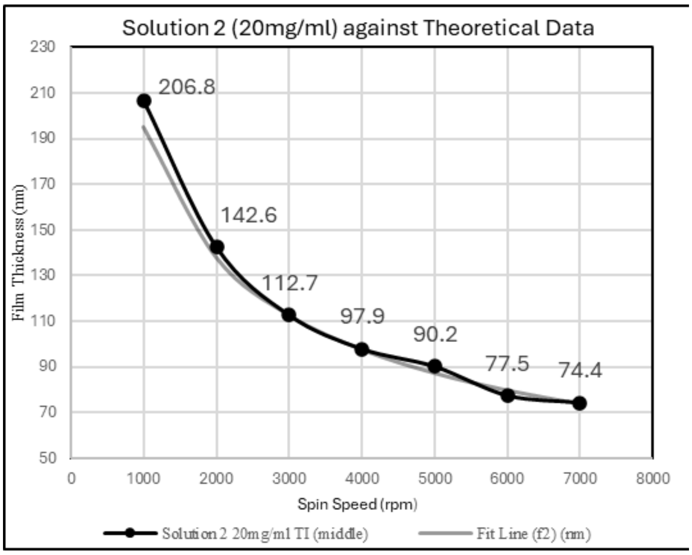
*Fig. 2. Effect of experimental factors compared to theoretical potential.*




$\boldsymbol{\mu_{f1}} = 54.1\ nm,\quad \boldsymbol{\sigma_{f1}} = 20.6\ nm,\quad \boldsymbol{RSD_{f1}} = 38.1\%,$

$\boldsymbol{\mu_{\%\ error1}} = 9.7\%,\quad \boldsymbol{RSD_{1000rpm}} = 6.2\%,\quad \boldsymbol{RSD_{7000rpm}} = 18.4\%$

*Fig. 3. Effect of experimental factors compared to theoretical potential.*




$$\boldsymbol{\mu_{f2}} = 112\ nm,\quad \boldsymbol{\sigma_{f1}} = 43.0\ nm,\quad \boldsymbol{RSD_{f2}} = 38.4\%,$$

$$\boldsymbol{\mu_{\%\ error2}} = 2.7\%,\quad \boldsymbol{RSD_{c+s/1000rpm}} = 34.3\%$$

$$\boldsymbol{RSD_{s/3000rpm}} = 29.4\%,\quad \boldsymbol{RSD_{c/7000rpm}} = 12.6\%$$

Relative standard deviation (RSD) of *sol. 1.* at *7000rpm* (corner measurements) compared to RSD of *sol. 1* at 1000rpm (corner measurements) shows recognisable differences *(12.2%)*. RSD of *sol. 2* at *1000 rpm* compared to *7000 rpm* shows the opposite relationship previously mentioned for *sol. 1. (-21.7%)*. This may indicate towards the effect of surface tension on the edge of a flat surface against the centrifugal force. Uniformity is directly proportional to an optimal margin of spin speed and viscosity. RSD sol. 1 *1000 rpm (6.2%)* compared to RSD sol. 2 *1000 rpm (34.3%)* further emphasises this.

The mean percentage error of *sol. 1 (9.7%)* and *sol. 2 (2.7%)* can show that lower viscosity solutions exacerbate experimental errors.

Using mean percentage error of *sol. 2* at a significantly low *(2.7%)* we can observe that film thickness is inversely proportional to the square root of spin speed.

The side thickness measurement of *1000 rpm* is *233nm* yet corner 1 and 2 reading is *175nm* and *93nm*. An *(88%)* variation between corners and (*150%)* difference between the side and minimum corner value. Which can be attributed to surface tension, evaporation and other mechanisms; The substrate is not circular, the sides provide more support for surface tension than the corners. This means more solution is flung off the substrate at the corners due to centrifugal force during spinning. Furthermore, air resistance will be higher at the corners than the sides; evaporation takes effect at a faster rate. Therefore, the film thickness will be relatively low, reducing uniformity.

Finally, for some samples the ellipsometer could not read a value most likely due to relatively extreme nonuniformity causing improper reflection of the laser.

## B. PECVD

Fig. 4. Film Thickness against x, y.

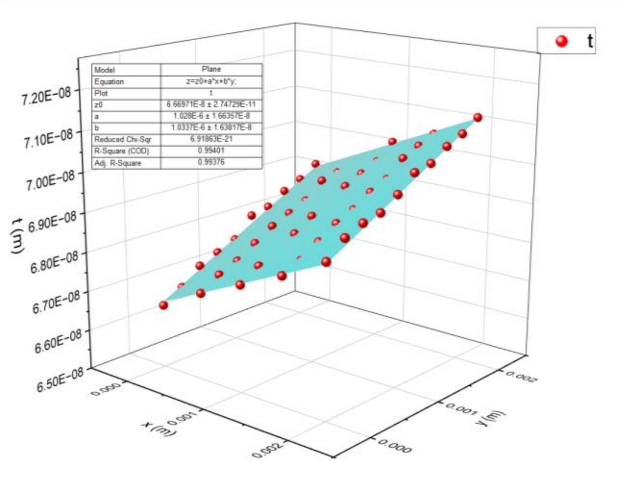



Fig. 5. Film thickness statistics.

1 Statistics on Columns (29/10/2025 16:42:48)

Notes

Input Data

Descriptive Statistics

| | N total | Mean | Standard Deviation | Sum | Minimum | Median | Maximum |
|---|---|---|---|---|---|---|---|
| t | 50 | 6.8888E-8 | 1.05262E-9 | 3.4444E-6 | 6.66E-8 | 6.88E-8 | 7.11E-8 |

Fig. 6. Refractive Index against x, y.

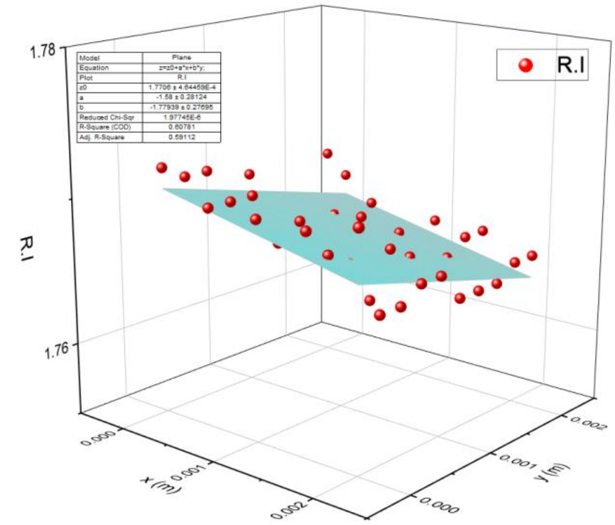



Fig. 7. Refractive Index Statistics

Statistics on Columns (29/10/2025 16:46:05)

Notes

Input Data

Descriptive Statistics

| | N total | Mean | Standard Deviation | Sum | Minimum | Median | Maximum |
|---|---|---|---|---|---|---|---|
| R.I | 50 | 1.76702 | 0.0022 | 88.351 | 1.764 | 1.766 | 1.772 |

Fig. 8. Thickness distribution Mu = mean Sigma = Standard Deviation

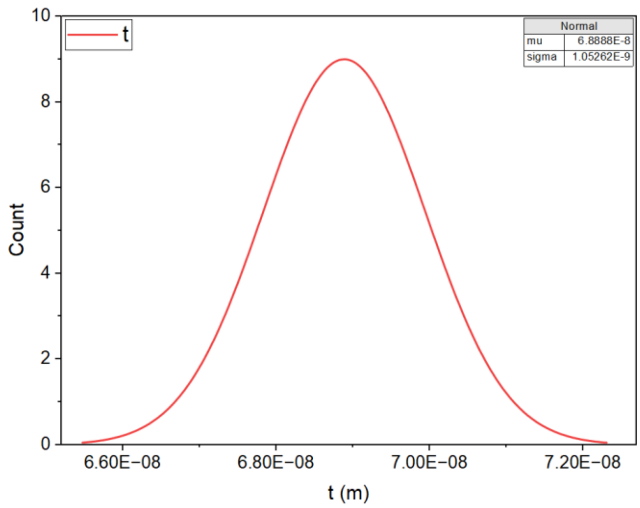
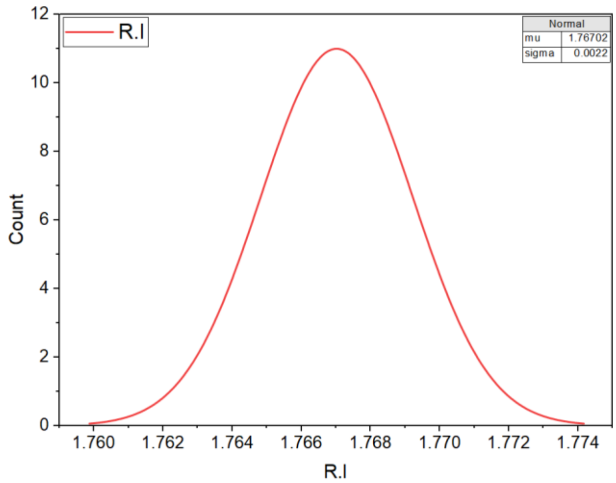



Fig. 9. Refractive Index distribution




The 3D surface plots *(fig. 4 & fig. 6)* display the spatial distribution of the film thickness and refractive index across the silicon wafer. Both datasets were fitted with a plane model to visualise overall uniformity. The thickness values range between $6.66 \times 10^{-8}\, m$ and $7.11 \times 10^{-8}$m, with a mean of $6.89 \times 10^{-8}\, m$ and a standard deviation of $1.05 \times 10^{-9}\, m$ (table in *Fig. 5*). This represents a very low variation *(≈1.5%)*, confirming excellent thickness uniformity across the wafer surface.

Similarly, the refractive index *(Fig. 6-9)* ranged from *1.764* to *1.772,* with a mean of *1.767* and standard deviation of *0.0022*, complementing *0.12%* variation. This extremely low deviation displays high optical uniformity and consistency in film composition.

Fig. 8 & Fig. 9 further demonstrate that thickness and refractive index follow a **normal distribution**, indicating towards experimental noise rather than process error.

In contrast, spin-coating provides much higher variability in the films, with relative standard deviations exceeding *(30-40%)*. This highlights the superior process control and uniformity available using PECVD compared to the manual process of spin-coating.

## IV. Methodological Improvements

In the cleaning of impurities of the silicon substrate; the substrates target areas were in contact with one another, this can cause incomplete reactions of the chemicals to the silicon substrate's surface. generating reduced uniformity of the later applied coating. – To improve this; a rack which holds the substrate can be incorporated to ensure even cleaning of the target area in the chemical bath.

The size and shape of each substrate varied. This can cause improper distribution and evaporation of the solution; one cause is surface tension of different shaped edges. To avoid this, uniformly shaped substrates reduce this error effect.

Nitrogen gun did not remove all the moisture, could bake the substrates instead. This also allows for slower evaporating solutions to be used more effectively since baking ensures even evaporation.

## V. Conclusion

PECVD displays much stronger uniformity and thickness accuracy. Spin-coating is advantageous due to its replicability using relatively affordable equipment. Furthermore, spin-coating has several experimental variables that must be optimised to ensure relative uniformity such as solution viscosity, spin speed, evaporation rate, cleaning process and static/dynamic spin coating among others. PECVD is a much more controlled and industrial technique allowing for replicability of much more accurate thin films than spin-coating.

## VI. References

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